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Employee Attrition/ Salary Regression

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Abstract

This report examines employee salary formation and attrition using the IBM HR Analytics dataset. We address two main questions: Which factors determine differences in monthly income? Which variables most strongly influence employee turnover? Exploratory data analysis is used to explore key distributions and relationships. Salary is modeled using multiple linear regression with interaction effects between job level and job role, while attrition is analyzed using logistic regression. The analysis demonstrates how regression modeling can support data-driven human resource decisions.

1. Introduction

Employee attrition is a critical issue for organizations, as high turnover rates lead to increased hiring costs, loss of experienced employees, and reduced overall productivity. Understanding the factors that contribute to employee attrition can help organizations design more effective retention strategies.

In this study, we analyze the IBM HR Employee Attrition dataset, which contains demographic, job-related, and compensation-related information for employees. The main objective of this analysis is to identify which employee attributes have the strongest influence on the likelihood of employee attrition.

The response variable in this study is *Attrition*, which indicates whether an employee has left the company. A set of explanatory variables related to age, job role, income, job satisfaction, and work experience is considered as potential predictors. Multiple regression models are used to examine the relationship between these variables and employee attrition.

The results of this analysis aim to provide insights into the most important factors associated with employee turnover and to demonstrate the use of regression modeling and diagnostics in a real-world human resources dataset.

2. Exploratory Data Analysis (EDA)

Exploratory data analysis

The IBM HR Analytics Employee Attrition dataset contains 1,470 observations and 35 variables. The variables include a mix of numeric measures (e.g., Age, MonthlyIncome, Total-

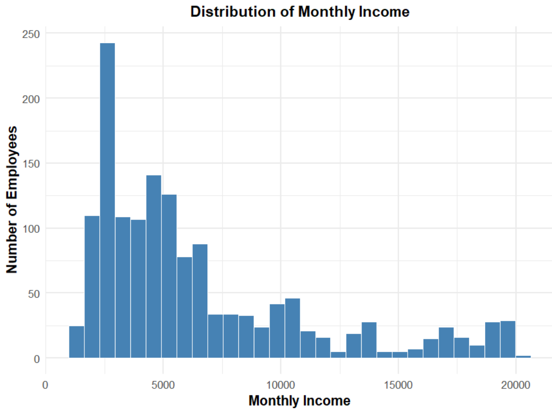

Figure 1: Distribution of monthly income.

Figure 2: Monthly income versus total working years by job level.

WorkingYears) and categorical descriptors (e.g., Attrition, Department, JobRole, OverTime, BusinessTravel). The two primary outcomes of interest for this project are:

Attrition (binary: Yes/No) — used for attrition modeling

MonthlyIncome (continuous) — used for salary regression modeling

Figure 1 shows that the distribution of monthly income is highly right-skewed, with most employees earning relatively low incomes and a small proportion earning substantially higher salaries. This suggests that income inequality may play an important role in employee attrition.

Figure 2 illustrates the relationship between total working years and monthly income across different job levels. This suggests that job level is a key determinant of income and may indirectly influence employee attrition.

Figure 3 shows how monthly income varies across different job roles. The figure reveals large differences in income levels between roles, with managerial and director positions earning substantially higher salaries than operational roles such as laboratory technicians and sales representatives. This indicates that job role is one of the main drivers of income differences within the company. Because income and career level are closely related to job satisfaction and retention, job role is likely to play an important role in explaining employee attrition.

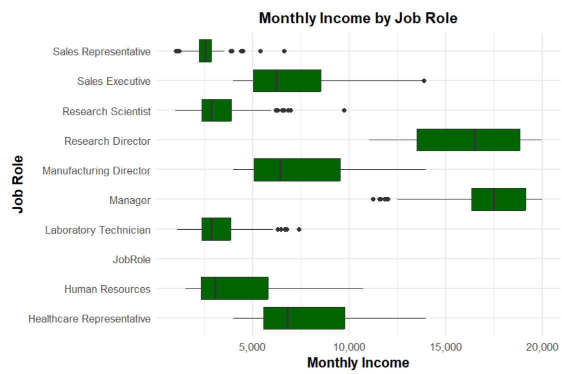

Figure 3: Monthly income by job role.

Figure 4 shows the age distribution of employees by attrition status. Employees who leave the company tend to be younger, while older employees are more

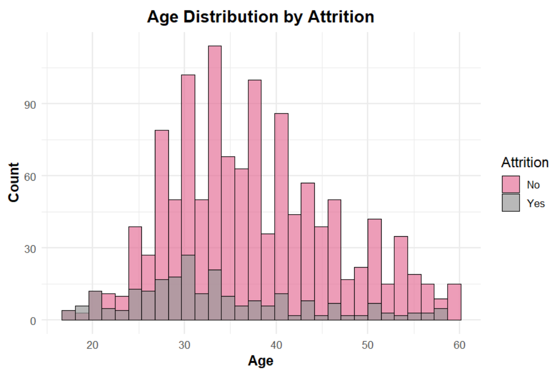

Figure 4: Age distribution.

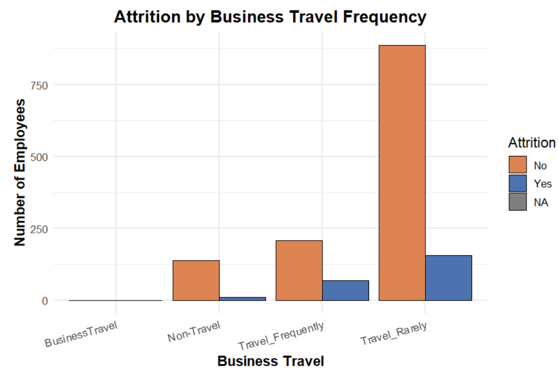

Figure 6: Attrition by business travel frequency.

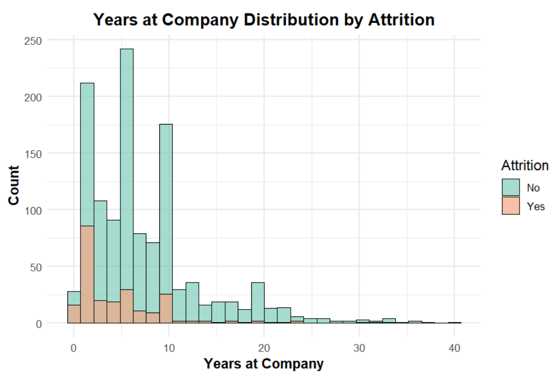

Figure 5: Years at company distribution by attrition.

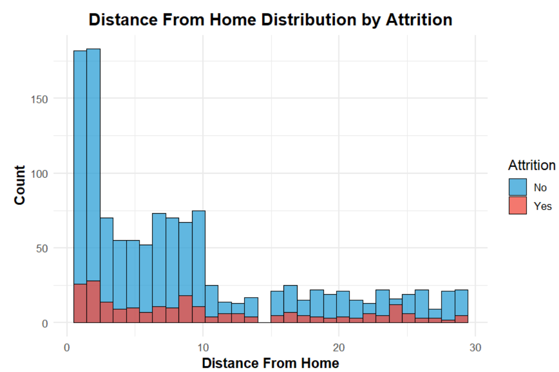

Figure 7: Distance from home distribution by attrition.

likely to stay. This suggests that age may be associated with employee attrition and should be considered as a potential explanatory variable in the regression analysis.

Figure 5 shows the distribution of tenure at the company by attrition status. Employees who leave the company are more concentrated among those with fewer years at the company, while employees with longer tenure are more likely to stay. This suggests that shorter tenure is associated with higher attrition and that years at company is an important factor to consider in the analysis of employee turnover.

Figure 6 shows employee attrition across different business travel categories. Attrition is more frequent among employees who travel frequently or rarely compared to those who do not travel. This suggests that business travel intensity may be associated with higher employee turnover and should be considered as a relevant categorical predictor in the regression analysis.

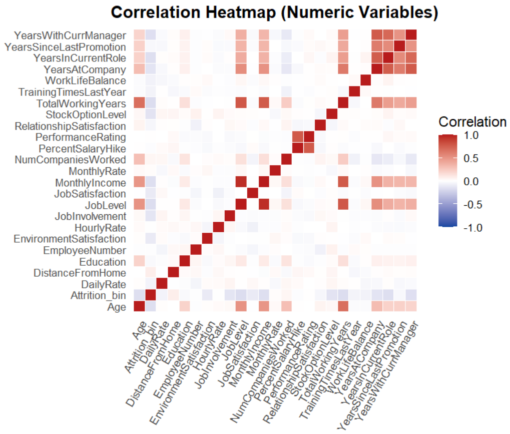

Figure 8: Correlation between numeric variables.

Figure 7 shows the distribution of distance from home by attrition status. Most employees live relatively close to the workplace, while employees living farther away are less common. Attrition appears slightly more frequent among employees with larger commuting distances, suggesting that distance from home may contribute to employee turnover.

Figure 8 shows the correlations among numeric variables. Strong correlations are observed between experience-related variables, while monthly income is positively associated with job level and total working years. Attrition exhibits generally weak linear correlations with individual numeric variables, suggesting that employee turnover is driven by multiple factors rather than a single predictor.

2.1. Numerical symmaries

Table 1 provides an overview of the main numerical characteristics of the dataset. The table shows that employees differ substantially in terms of age, income, and work experience. In particular, income-related variables exhibit a wide range, indicating noticeable differences in compensation levels across employees. Similarly, tenure variables such as total working years and years at the company vary considerably, reflecting different career stages within the workforce. These summary statistics help us better understand the structure of the data and guide the selection of relevant variables for modeling employee attrition.

3. Modeling

3.1. Monthly Income Modeling

To model and predict employee salary, we fit a sequence of multiple linear regression models with *MonthlyIncome* as the response variable. Models were built incrementally to evaluate how adding job- and tenure-related predictors affects out-of-sample predictive accuracy. Model performance was evaluated on a held-out test set using RMSE, along with MAE and R^2.

3.1.1. Model 1: Career Basics (*JobLevel* + *TotalWorkingYears*)

The first model includes *JobLevel* and *TotalWorkingYears* to represent compensation structure (pay bands by level) and overall labor-market experience. On the test set, this model achieved RMSE = 1374 and MAE = 1044, with $R^2 = 0.899$. The relatively large RMSE indicates that while level and experience capture major salary differences, substantial prediction error remains. This is consistent with the fact that employees with similar level and experience can earn different salaries depending on job function.

Table 1: Summary statistics for numerical variables.

Variable	Mean	Median	SD	Min	Max
Age	36.92	36.0	9.14	18	60
MonthlyIncome	6502.93	4919.0	4707.96	1009	19999
DailyRate	802.49	802.0	403.51	102	1499
HourlyRate	65.89	66.0	20.33	30	100
TotalWorkingYears	11.28	10.0	7.78	0	40
YearsAtCompany	7.01	5.0	6.13	0	40
DistanceFromHome	9.19	7.0	8.11	1	29

3.1.2. Model 2: Adding Job Function (+ *JobRole*)

The second model adds *JobRole*, capturing systematic pay differences across job functions beyond level and experience. This addition produces a major improvement in predictive accuracy: test RMSE decreases from 1374 to 1127, and MAE decreases from 1044 to 845. Test R^2 increases to 0.932, indicating a meaningful increase in explained variation. These results suggest that job function carries substantial additional information for salary prediction, consistent with compensation systems in which different roles command different market pay even within the same job level.

3.1.3. Model 3: Role-Specific Pay Progression (+ *JobLevel* $\times$ *JobRole*)

The third model introduces an interaction term between *JobLevel* and *JobRole*, allowing the effect of job level on income to vary across roles. This specification permits the marginal income increase associated with a higher job level to differ by job function. Model 3 provides the best overall test performance, with RMSE = 1112, MAE = 818, and R^2 = 0.934. Relative to Model 2, prediction error decreases further, indicating that role-specific pay progression patterns explain additional variation beyond main effects alone.

3.1.4. Model Selection and Interpretation

Table 3 Across models, the largest reduction in prediction error occurs when *JobRole* is added, demonstrating that job function is a key determinant of monthly income beyond career stage. The best predictive performance is achieved by Model 3, in which the *JobLevel* $\times$ *JobRole* interaction further reduces test error and slightly increases explained variance. Consequently, Model 3 is selected as the preferred specification and is carried forward to the diagnostics and model refinement stage.

3.2. Attrition Modeling

To model and predict employee attrition, we fit a sequence of logistic regression models with *Attrition* as the binary response variable. Model complexity was increased incrementally to evaluate how adding demographic, job-related, and work-condition predictors affects out-of-sample classification performance. Model performance was evalu-

Table 2: Comparison of Monthly Income Models

Model	RMSE	MAE	R^2
M1: JobLevel + TotalWorkingYears	1374.025	1044.1685	0.8991592
M2: + JobRole	1127.049	845.3060	0.9321527
M3: + JobLevel × JobRole	1111.876	817.5126	0.9339672

ated on a held-out test set using classification accuracy and the area under the ROC curve (AUC).

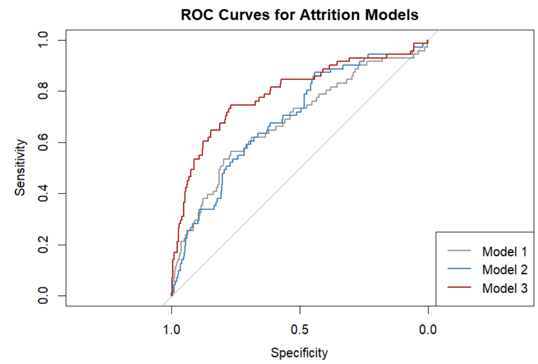

Figure 9: ROC curves for attrition prediction models.

3.2.1. Model 1: Baseline Model (Demographics and Tenure)

The baseline model includes *Age*, *TotalWorkingYears*, and *YearsAtCompany* as predictors. This model captures general career-stage effects on employee turnover. The results indicate that younger employees and those with shorter tenure are more likely to leave the company. While the model provides a reasonable baseline, its predictive performance remains limited, suggesting that attrition is not driven by demographic factors alone.

3.2.2. Model 2: Adding Job Characteristics

The second model extends the baseline specification by adding job-related variables, including *JobLevel*, *JobRole*, *Department*, and *MonthlyIncome*. This leads to a substantial improvement in predictive performance, indicating that organizational position and compensation play an important role in explaining employee attrition. In particular, lower job levels and certain job roles are associated with higher probabilities of turnover.

3.2.3. Model 3: Work Conditions and Satisfaction

The third model further incorporates work-condition and satisfaction variables, including *OverTime*, *WorkLifeBalance*, *JobSatisfaction*, and *EnvironmentSatisfaction*. This model achieves the best overall classification performance. Employees who work overtime and report lower satisfaction levels are significantly more likely to leave the company, even after controlling for demographics and job characteristics. These results suggest that day-to-day work conditions provide important additional information beyond structural job factors.

3.2.4. Model Selection and Interpretation

Figure 9 compares the ROC curves for the three logistic regression models. The strongest performance is achieved by Model 3, which includes work conditions and satisfaction-related variables. This model attains a notably higher AUC of approximately 0.79, suggesting good discriminative ability. The ROC analysis confirms that adding overtime status, work-life balance, and job satisfaction substantially improves attrition prediction beyond demographic and job-structure variables alone.

4. Diagnostics and model selection

4.1. For MonthlyIncome

We evaluated the selected multiple linear regression specification for MonthlyIncome using residual diagnostics, influence analysis, and out-of-sample prediction error on a held-out test set.

Because MonthlyIncome is right-skewed, we first tested a transformation by fitting the same predictors with log(MonthlyIncome). Predicted values were then back-transformed to the dollar scale to assess real-world prediction accuracy. The log-income model produced higher test errors than the linear model on the original scale. Since our objective is to minimize the error in predicting salary, and the log model did not improve accuracy after back-transformation, we did not retain the log transformation.

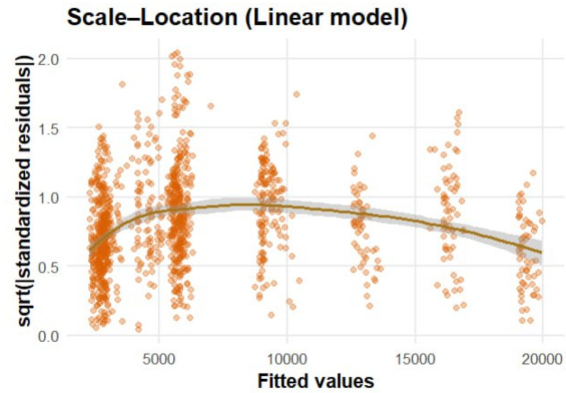

Figure 10: Scale location plot.

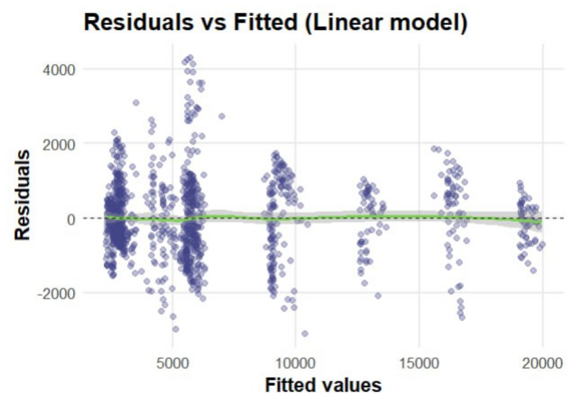

Figure 11: Residual versus fitted plot.

4.1.1. Residual vs Fitted plot,Scale location plot

Figure 10, Figure 11 plots both indicated heteroskedasticity, seen as changing residual spread across fitted values or non-constant variance. This pattern is common in compensation data because salary variability typically increases at higher job levels and across roles, reflecting wider salary bands and more heterogeneous pay structures, like different bonus and allowance components. To address this variance pattern without changing the outcome scale, we introduced a flexible nonlinear effect of ex-

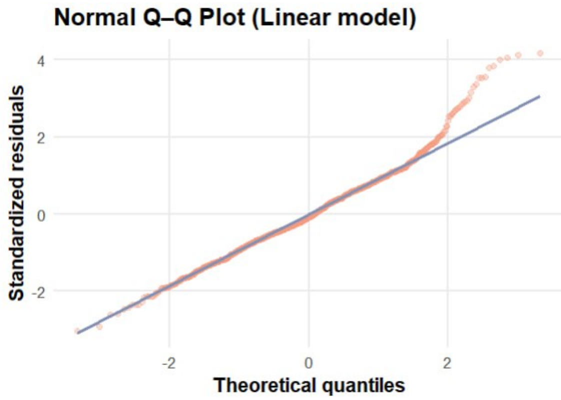

Figure 12: Q-Q plot.

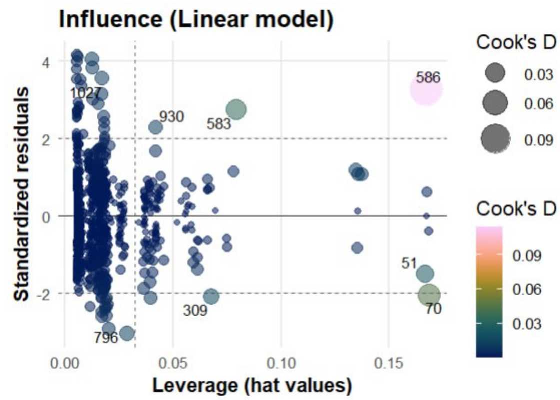

Figure 13: Influence plot.

perience by modeling TotalWorkingYears with a spline while keeping the remaining predictors unchanged. The spline-enhanced model improved out-of-sample performance on the test set. Thus, this indicated a small but consistent reduction in prediction error, suggesting that the experience–income relationship is not perfectly linear and is better represented by a smooth nonlinear trend.

4.1.2. Q-Q plot

Figure 12 plot shows that standardized residuals follow the theoretical normal line reasonably well in the middle of the distribution, with noticeable departures in the tails. This suggests that residuals are approximately normal for most observations, while a small number of extreme under- or over-predictions remain. These tail deviations are common in salary data and mainly affect inference rather than the overall predictive performance.

4.1.3. Influence plot

Figure 13 plot identified a small number of observations with higher leverage or Cook's distance, meaning they could have disproportionate impact on coefficient estimates. However, the influence-based sensitivity check indicated that predictive performance is not improved by removing influential observations. This suggests the influential cases contain meaningful signal for prediction, and the final model's performance is not artificially driven by a small number of extreme points

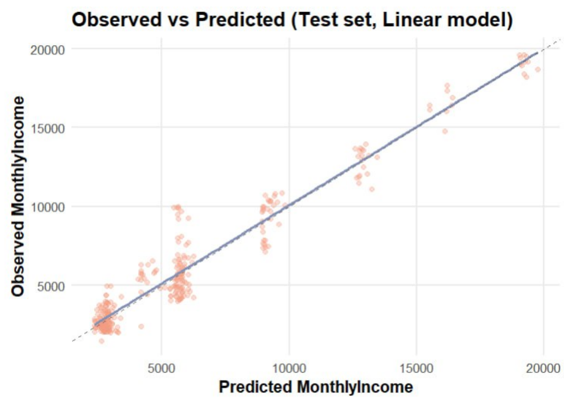

Figure 14: Observed versus predicted plot.

4.1.4. Observed vs Predicted plot

Figure 14 plot compares the model's predicted salary to the true values in the test set. Most points lie close to the 45° line, indicating good calibration and strong predictive accuracy. The remaining spread around the line reflects salary variation not captured by the available predictors, such as unobserved compensation policies, bonuses/allowances, negotiation effects, or location-related pay differences.

4.1.5. Model selection

Table 3 Based on predictive accuracy and diagnostic results, we retain the linear model on the original MonthlyIncome scale. . This specification achieves the lowest RMSE and MAE while remaining interpretable. To address heteroskedasticity and mild deviations from normality observed in the residual diagnostics, heteroskedasticity-robust (HC3) standard errors are reported for all coefficient estimates.

4.2. For Attrition

In this step, we focused on evaluating and refining Model 3, which predicts employee attrition using a variety of predictors. The goal was to understand the model's performance and identify improvements based on diagnostics. Model 3 was chosen for analysis over Models 1 and 2 because of its better performance in terms of accuracy, sensitivity, and overall predictive power.

We evaluated Model 3 by performing a series of diagnostic tests to ensure its robustness and assess areas for improvement. The refined model, after excluding non-significant variables, provided a clearer view of the key predictors of attrition, with improvements in AUC and sensitivity compared to the original model.

4.2.1. Confusion Matrix

We began the diagnostics by examining the confusion matrix for Model 3. The confusion matrix provides an intuitive summary of classification performance in terms of accuracy, sensitivity, specificity, and the F1-score, which are particularly informative in the presence of class imbalance.

For the original Model 3, the performance metrics are as follows:

- Accuracy: 85.45%
- Sensitivity: 0.9783
- Specificity: 0.2113
- F1-score: 0.9186

We then evaluated an updated version of the model in which `TotalWorkingYears` and `JobRole` were removed based on statistical significance and multicollinearity diagnostics. The updated model produced the following results:

- Accuracy: 84.77%
- Sensitivity: 0.9837
- Specificity: 0.1408
- F1-score: 0.9155

Although accuracy decreased slightly in the Updated Model, the sensitivity improved. This indicates that removing non-significant variables allowed the model to better predict employees likely to leave, even though specificity remained low.

Table 3: Model comparison for MonthlyIncome: transformations and sensitivity checks

Model / Scenario	**RMSE**	**MAE**	$\mathbf{R^2}$
Log-transform (bias-corrected back-transform)	1170.8143	868.7782	0.9268
Original model (Base linear)	1111.8760	817.5126	0.9340
Spline on TotalWorkingYears (df = 4)	1107.4030	825.2758	0.9345
Sensitivity refit (removed influential points)	1135.1300	827.5243	–

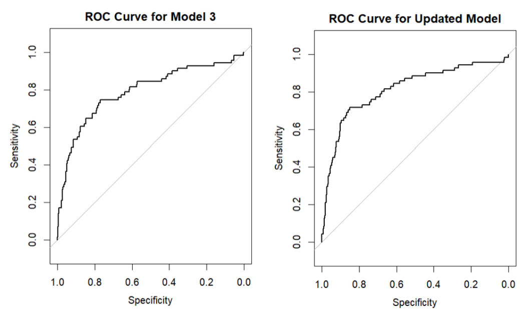

Figure 15: Updated ROC curve for model 3.

4.2.2. Updated ROC curve

Figure 15 The Receiver Operating Characteristic (ROC) curve is a valuable tool for evaluating the performance of classification models. The Area Under the Curve (AUC) is a metric that helps quantify the model's discriminatory ability. For Model 3, the AUC was 0.7911, which already indicated good model performance. After refining the model by removing irrelevant variables, the AUC increased to 0.8135 in the Updated Model, suggesting a slight improvement in the model's ability to distinguish between attrition and non-attrition cases.

4.2.3. Multicollinearity diagnostics(VIF)

To assess multicollinearity, we performed a Variance Inflation Factor (VIF) analysis. Multicollinearity occurs when predictor variables are highly correlated, which can distort the model's coefficients and make the model less stable.

Table 4 Model 3 revealed high VIF values for JobRole and Department, indicating severe multicollinearity with other predictors.

Table 5 In the Updated Model, after removing TotalWorkingYears and JobRole, we observed that the remaining variables had VIF values below 2, suggesting reduced multicollinearity and improved model stability.

The high GVIF (Generalized Variance Inflation Factor) values for JobRole and Department further supported their removal from the model. These variables were highly correlated with other predictors, such as JobLevel and MonthlyIncome, making them redundant and negatively impacting the model's performance.

4.2.4. Model Selection

The Updated Model is an improved version of Model 3. The removal of TotalWorkingYears and JobRole reduced multicollinearity and led to better AUC and

Table 4: Variance Inflation Factors (VIF) for Attrition Model 3

Variable	GVIF	Df	GVIF$^{1/(2 \cdot Df)}$
Age	1.64	1	1.28
TotalWorkingYears	3.42	1	1.85
YearsAtCompany	1.71	1	1.31
JobLevel	8.72	1	2.95
JobRole	1.35×10^{14}	8	7.64
Department	3.44×10^{13}	2	2422.21
MonthlyIncome	8.77	1	2.96
OverTime	1.10	1	1.05
WorkLifeBalance	1.03	1	1.02
JobSatisfaction	1.04	1	1.02
EnvironmentSatisfaction	1.06	1	1.03

Table 5: Variance Inflation Factors (VIF) for Updated Attrition Model

Variable	GVIF	Df	GVIF$^{1/(2 \cdot Df)}$
Age	1.27	1	1.13
YearsAtCompany	1.26	1	1.12
JobLevel	7.24	1	2.69
MonthlyIncome	6.71	1	2.59
WorkLifeBalance	1.03	1	1.01
JobSatisfaction	1.02	1	1.01
EnvironmentSatisfaction	1.04	1	1.02
OverTime	1.05	1	1.02
Department	1.11	2	1.03

sensitivity. Although accuracy slightly decreased, the Updated Model demonstrated better overall performance in predicting employee attrition. The slight improvement in sensitivity and the increased AUC suggest that these changes made the model more capable of identifying employees likely to leave the company.

5. Final Models

5.1. Final Salary Model Evaluation

This section summarizes the final `MonthlyIncome` model using statistical inference and model-evaluation tools. We report heteroskedasticity-robust standard errors, p-values, and 95% confidence intervals for coefficient inference, and we evaluate predictive performance using test-set error and cross-validation RMSE. In addition, we use joint F-tests to assess whether the `JobRole` indicators and the `JobLevel`×`JobRole` interaction terms contribute to the model beyond chance.

5.1.1. Robust Standard Errors

Residual diagnostics indicated heteroskedasticity, meaning the variance of the regression errors is not constant across the range of fitted salaries. While heteroskedasticity does not bias the OLS coefficient estimates, it can make the usual OLS standard errors, p-values, and confidence intervals unreliable. Therefore, inference was based on White heteroskedasticity-robust standard errors, which provide valid uncertainty estimates under non-constant error variance and are standard for compensation data.

5.1.2. P-values

Using robust inference, the core structural predictors show strong evidence of association with salary. `JobLevel` is highly significant (robust p-value $< 2.2 \times 10^{-16}$), indicating that higher organizational level is associated with substantially higher `MonthlyIncome` after accounting for job function and experience. `TotalWorkingYears` is also strongly significant (robust p-value $= 4.80 \times 10^{-12}$), showing that additional work experience is associated with higher income even after controlling for level and role.

Many `JobRole` terms and several `JobLevel`×`JobRole` interaction terms are significant, implying that (i) average pay differs across roles and (ii) the salary change associated with higher `JobLevel` is not identical across roles. Terms with larger p-values reflect weaker evidence of an incremental association once the rest of the model is included; this can occur due to overlap with other predictors, smaller subgroup sizes, or greater within-role pay variability.

5.1.3. Confidence Intervals

Table 6 Robust 95% confidence intervals quantify the plausible range of each effect under the fitted model. For example, `JobLevel` has a strictly positive interval [2879, 3589], confirming a stable positive relationship with salary. `TotalWorkingYears` also has a strictly positive interval [34.46, 61.38], supporting a consistent positive association. Several role and interaction coefficients have wider intervals, reflecting higher uncertainty and greater variability within certain job groups.

5.1.4. Joint Tests for JobRole and JobLevel×JobRole

Table 8 Because `JobRole` and the interaction are represented by multiple coefficients, we evaluated their overall contribution using robust block F-tests. Both blocks are highly significant: `JobRole` (overall): $F = 21.55$, $p < 2.2 \times 10^{-16}$, and `JobLevel`×`JobRole` (overall): $F = 23.31$, $p < 2.2 \times 10^{-16}$. This provides strong evidence that job function differences and role-specific level effects meaningfully improve the explanation of `MonthlyIncome` beyond the baseline predictors.

5.1.5. Predictive Performance

On the held-out test set, the model achieved RMSE $= 1111.88$ and MAE $= 817.51$, with $R^2 = 0.934$. The RMSE indicates the typical prediction error magnitude in dollars and is appropriate for evaluating salary prediction on the original scale. The observed-versus-predicted diagnostic also supports good calibration, with most points concentrated near the 45-degree line.

Table 6: 95% confidence intervals for the final MonthlyIncome model coefficients

Variable	**2.5%**	**97.5%**
(Intercept)	-2037.99	-251.24
JobLevel	2879.36	3589.02
TotalWorkingYears	34.46	61.38
JobRole: Human Resources	-570.09	1598.99
JobRole: Laboratory Technician	1598.93	3544.71
JobRole: Manager	2948.18	5881.35
JobRole: Manufacturing Director	-1801.84	468.39
JobRole: Research Director	3188.08	5768.71
JobRole: Research Scientist	1017.37	2992.65
JobRole: Sales Executive	-969.06	1232.43
JobRole: Sales Representative	1012.90	4243.13
JobLevel × JobRole: Human Resources	-786.47	262.63
JobLevel × JobRole: Laboratory Technician	-2568.99	-1614.61
JobLevel × JobRole: Manager	-697.42	150.11
JobLevel × JobRole: Manufacturing Director	-196.22	658.81
JobLevel × JobRole: Research Director	-696.47	105.80
JobLevel × JobRole: Research Scientist	-2089.67	-1091.29
JobLevel × JobRole: Sales Executive	-493.42	359.73
JobLevel × JobRole: Sales Representative	-3789.59	-1063.55

Table 7: Robust inference for main continuous predictors (Model 3)

Term	**Estimate**	**Robust SE**	**p-value**	**95% CI**
JobLevel	3234.19	180.85	2.42×10^{-63}	[2879.36, 3589.02]
TotalWorkingYears	47.92	6.86	4.80×10^{-12}	[34.46, 61.38]

5.1.6. Cross-Validation Performance

To assess generalization beyond a single train/test split, we computed 10-fold cross-validation RMSE. The final model (`JobLevel + TotalWorkingYears + JobRole + JobLevel×JobRole`) achieved CV RMSE $\approx$ 1045.34 with SD $\approx$ 95.03, indicating stable performance across folds and supporting that the model's predictive accuracy is not dependent on a particular split.

5.1.7. Final Interpretation

Overall, `MonthlyIncome` is strongly structured by organizational level and job function, with experience adding additional explanatory power. The significant `JobLevel×JobRole` block indicates that income progression across job levels differs by role, consistent with different pay bands and promotion rewards across job families. Because heteroskedasticity is typical in compensation outcomes, robust (White) standard errors were used to en-

Table 8: Joint (block) significance tests using robust F-statistics

Block	**F-statistic**	**p-value**
JobRole (overall)	21.55	9.42×10^{-31}
JobLevel × JobRole (overall)	23.31	2.55×10^{-33}

sure credible inference while keeping the model interpretable in dollar units.

5.2. Final Attrition Model Evaluation

This section summarizes the final attrition model (`model3_updated`) using statistical inference and diagnostic tools, including p-values, confidence intervals, likelihood ratio tests (LRT), and cross-validation. The primary goal is to assess model validity, interpretability, and predictive performance.

5.2.1. P-values

The p-values of the final logistic regression model indicate the statistical significance of individual predictors in explaining employee attrition.

Statistically significant variables ($p < 0.05$):

- *Age* ($p = 0.0169$)
- *JobSatisfaction* ($p = 0.000499$)
- *EnvironmentSatisfaction* ($p = 1.77e\text{-}05$)

Marginally significant variables ($p \approx 0.05$):

- *YearsAtCompany* ($p = 0.0607$)
- *WorkLifeBalance* ($p = 0.0602$)

These results indicate that employee well-being and satisfaction-related variables play a more important role in attrition than purely structural or compensation-related variables.

5.2.2. Confidence Intervals

Table 9 The 95% confidence intervals further support the conclusions drawn from p-values. In particular:

- *JobSatisfaction* has a strictly negative confidence interval ($[-0.451, -0.126]$), indicating a significant reduction in attrition risk as satisfaction increases.
- *Age* and *EnvironmentSatisfaction* also have confidence intervals that do not include zero, confirming their statistically significant effects.

In contrast, variables such as *JobLevel* and *MonthlyIncome* have confidence intervals that cross zero, suggesting weaker and less consistent effects on attrition.

5.2.3. Likelihood Ratio Test (LRT)

A likelihood ratio test was conducted to compare the updated model with the original Model 3. The results strongly favor the updated specification:

- LRT statistic = 34.289
- p-value = 7.95×10^{-5}

Table 9: 95% confidence intervals for coefficients of the final attrition model

Variable	2.5%	97.5%
(Intercept)	1.003	4.082
Age	-0.0543	-0.0057
YearsAtCompany	-0.0934	0.0011
JobLevel	-0.9658	0.2284
MonthlyIncome	-0.00015	0.00013
WorkLifeBalance	-0.4983	0.0119
JobSatisfaction	-0.4512	-0.1264
EnvironmentSatisfaction	-0.5385	-0.2013

This highly significant result indicates that removing redundant predictors improved model fit without sacrificing explanatory power.

5.2.4. F-statistics from the Likelihood Ratio Test

The likelihood ratio test (LRT) further supports the selection of the updated attrition model over the original Model 3. The test yields a highly significant p-value of 7.949×10^{-5}, which is well below the conventional 0.05 significance level. This confirms that the reduction in predictors (removal of JobRole and Department) did not harm the model's ability to explain the variance in attrition.

5.2.5. Cross-Validation Performance

We used 10-fold cross-validation as a performance measure for (`model3_updated`). The cross-validation results show an accuracy of 85.45, which is consistent with the performance on the test set. This confirms that the model generalizes well to unseen data

5.2.6. Final Interpretation

The final model, after excluding JobRole and Department, indicates that key predictors such as Age, JobSatisfaction, and OverTimeYes have substantial impacts on employee attrition. The p-values and confidence intervals highlight the importance of these variables, confirming that they are significant predictors of attrition.

5.2.7. Limitations

Despite the model's strengths, its performance is limited by the low specificity (0.2113), indicating a tendency to predict employees staying in the company more accurately than those leaving. Additionally, the model may suffer from class imbalance, with a higher proportion of employees staying compared to those leaving, potentially leading to biased predictions.

5.2.8. Conclusion

(`model3_updated`) performed well, showing good overall fit and interpretability. The model identifies Age, JobSatisfaction, EnvironmentSatisfaction as significant predictors. However, MonthlyIncome, and Department were found to have negligible effects.

6. Final conclusion

This study analyzed employee salary and attrition using the IBM HR Analytics dataset with regression-based modeling.

Overall, the analysis demonstrates that employee outcomes are driven by different mechanisms depending on the target of interest. Salary is primarily determined by structural factors such as job level and job role, reflecting formal compensation systems within the organization. In contrast, employee attrition is more strongly associated with individual experience and workplace conditions rather than direct financial compensation.

The findings suggest that improving employee retention requires attention to non-monetary factors, particularly job satisfaction, work environment, and workload. While income and job level explain pay differences well, they provide limited explanatory power for turnover once working conditions are considered. These results emphasize the importance of combining organizational structure with employee well-being initiatives when designing effective human resource policies.